a

Spurious terms per equation

Terms per equation

50
20
10
5
2
1
0

10^2 $10^{2.5}$ 10^3 $10^{3.5}$ 10^4 $10^{4.5}$ 10^5

n

SNR = 49 dB

b

True terms missed per equation

Terms per equation

4
3
2
1
0

10^2 $10^{2.5}$ 10^3 $10^{3.5}$ 10^4 $10^{4.5}$ 10^5

n

c

Mean selection frequency of spurious terms

Selection frequency

1.00
0.75
0.50
0.25
0.00

10^2 $10^{2.5}$ 10^3 $10^{3.5}$ 10^4 $10^{4.5}$ 10^5

n

d

Terms per equation

50
20
10
5
2
1
0

10 20 30 40 50 60 ∞

SNR (dB)

n = 5000

e

Terms per equation

4
3
2
1
0

10 20 30 40 50 60 ∞

SNR (dB)

f

Selection frequency

1.00
0.75
0.50
0.25
0.00

10 20 30 40 50 60 ∞

SNR (dB)

Bayesian−ARGOS ARGOS SINDy